5G/6G COMMUNICATION-ECA5604

Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

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SLOT-**A**

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Objective – 1

Analyze Steering Angle (Degrees)

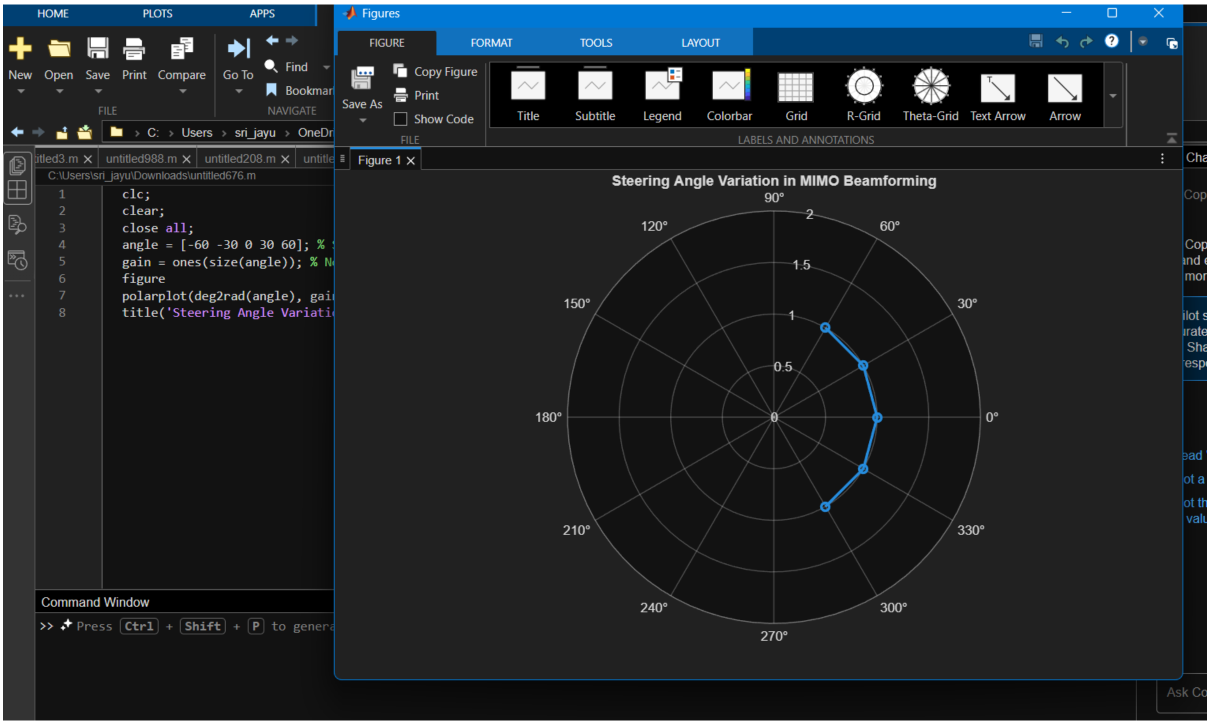
Analyze different steering angles of a MIMO beamforming system using MATLAB. Observe how
the beam can be directed toward various target directions by varying the steering angle.

MATLAB Code:

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = ones(size(angle)); % Normalized Gain
figure
```

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')



Objective – 2

Compare Array Gain (dB)

Compare the array gain obtained at different steering angles using MATLAB. Observe the improvement in signal strength achieved through beamforming.

MATLAB Code:

clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure

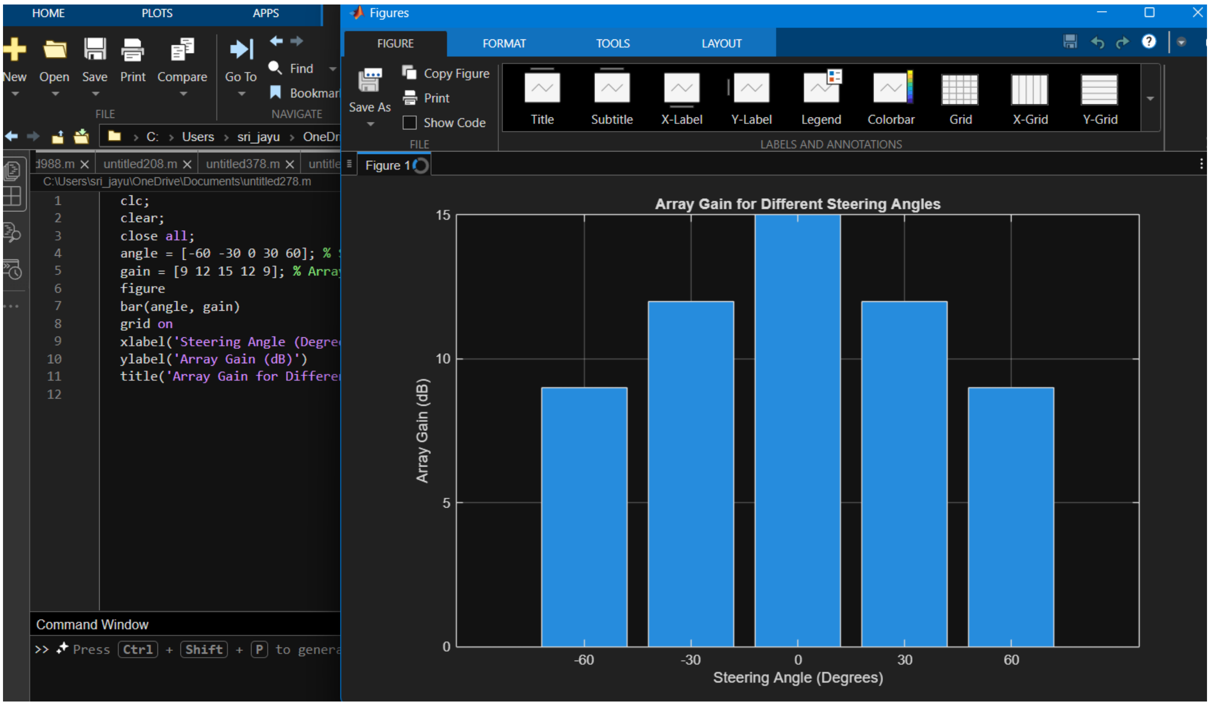
bar(angle, gain)

grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')



Objective – 3

Evaluate Beam Direction Error (Degrees)

Analyze the difference between the desired steering angle and the measured beam direction using

MATLAB. Compare the beam direction error for various steering angles and evaluate the beam

steering accuracy.

MATLAB Code:

clc;

clear;

close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)

actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)

error = abs(desired - actual); % Beam Direction Error

figure

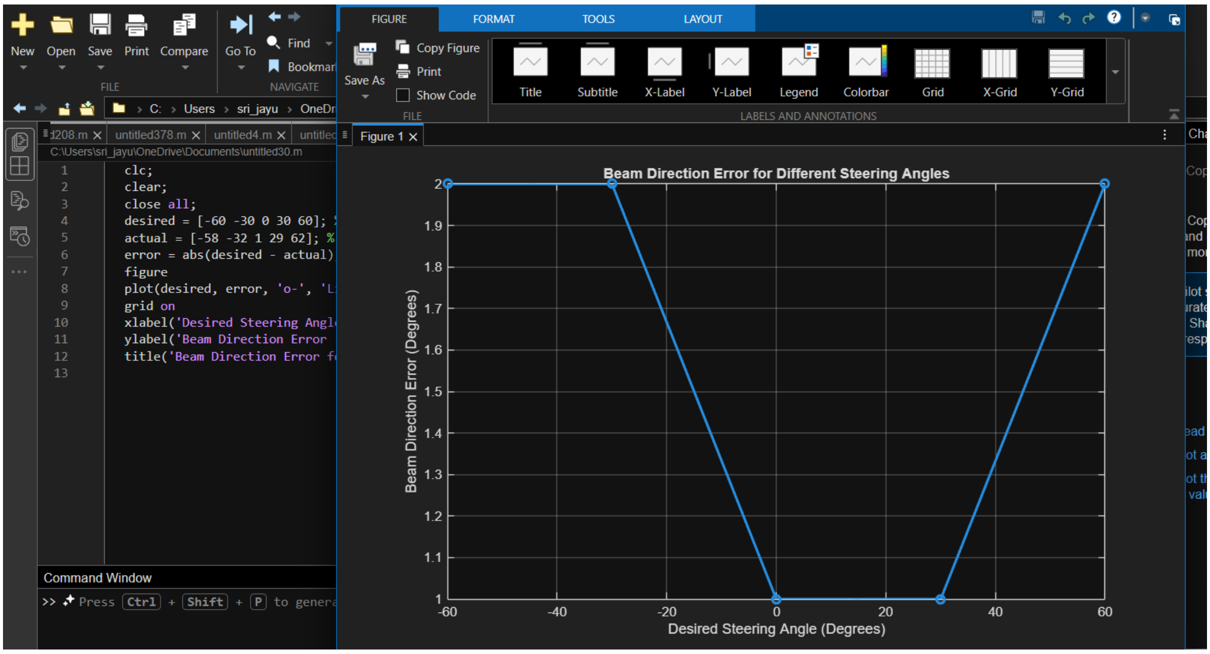
```
plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')
```



Result

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO

beamforming system. The results showed that beam steering effectively directs the transmitted

energy toward different angles, while maintaining high array gain and minimizing beam direction

error.